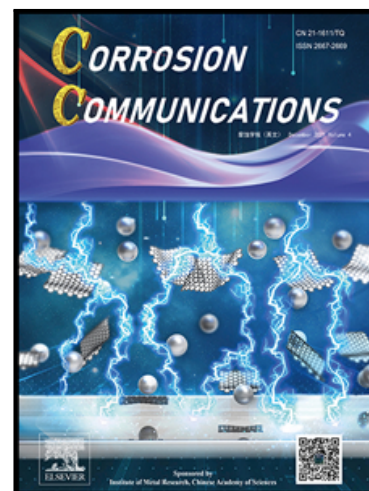


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Failure of a laser powder bed fusion fabricated GH3536 superalloy reverse-flow combustor flame tube under near-service conditions

Yuxuan Duan , Haohan Ni , Lei Qian , Xiaohong Shao , Qi Zeng ,
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Highlights:

- First study on the failure of a laser powder bed fusion fabricated GH3536 superalloy reverse-flow combustor flame tube under near-service conditions.
- Analysis of the reasons contributing to distinct crack modes at the primary hole edge and inter-hole area.
- Corrosion pits on the inner wall surface act as initiation sites for cracks.
- Multi-scale characterizations reveal grain boundaries as preferential paths for oxidation and sulfidation.
- Fatigue-creep, oxidation and hot corrosion synergistically accelerate crack initiation and propagation.

Research Article**Failure of a laser powder bed fusion fabricated GH3536 superalloy reverse-flow combustor flame tube under near-service conditions**

Yuxuan Duan^{a, 1}, Haohan Ni^{a, b, 1}, Lei Qian^c, Xiaohong Shao^b, Qi Zeng^{c, *}, Huajie Yang^{b, *}, Jiangwei Wang^{a, *}

^a Center of Electron Microscopy, State Key Laboratory of Silicon Materials, School of Materials Science and Engineering, Zhejiang University, Hangzhou 310027, China

^b Suzhou Laboratory, Suzhou 215123, China

^c AECC Hunan Aviation Powerplant Research Institute, Zhuzhou 412002, China

¹These authors contributed equally to this work.

*Corresponding authors.

E-mail addresses: 598630892@qq.com (Q. Zeng); yanghj@szlab.ac.cn (H.J. Yang); jiangwei_wang@zju.edu.cn (J.W. Wang).

Abstract

Additive manufacturing (AM), as a disruptive technology, has been increasingly applied in the aerospace industry due to its unique advantages in fabricating complex and precise parts. However, systematic investigations into the failure mechanisms of AM-fabricated aerospace parts under service conditions remain limited. This study investigates the failure mechanisms of an AM-fabricated GH3536 superalloy reverse-flow combustor flame tube subjected to near-service conditions. Results show that cracking on the inner wall surface of the flame tube is dominated by the synergistic effects of fatigue-creep, oxidation and hot corrosion, occurring predominantly in an intergranular mode with partial transgranular features. Corrosion behavior is characterized by a porous and heterogeneous corrosion layer, intergranular oxidation and sulfidation, and the formation of corrosion pits. At the primary hole edge, intragranular corrosion pits initiate transgranular cracks, which transition into intergranular propagation toward the inter-hole area, where corrosion products and pits along grain boundaries further facilitate the initiation and propagation of intergranular cracks.

Keywords: Additive manufacturing, Flame tube, Crack propagation, Corrosion, Failure mechanisms

1. Introduction

Additive manufacturing (AM) has been increasingly applied in the aerospace industry owing to its significant advantages over conventional manufacturing techniques, including shorter production cycles, greater design flexibility for customization and the capability to integrally fabricate complex structures [1-3]. In 2024, the global market for AM in aerospace applications was valued at approximately USD 5.4 billion and is projected to reach USD 19.6 billion by 2033. With the increasing application of micro aero-engines, the compact size, intricate structure and diverse materials of reverse-flow combustors (a critical component of aero-engines) have posed substantial challenges to conventional manufacturing techniques [4, 5]. Among various AM techniques, laser powder bed fusion (L-PBF) has been widely adopted for fabricating reverse-flow combustors [6, 7]. Currently, studies on the AM-fabricated aerospace parts largely focus on optimizing AM and post-treatment parameters to mitigate inherent printing defects and enhance material mechanical properties [8-10]. However, the structural evolution and reliability of AM-fabricated materials and parts under service conditions remain largely unclear. The flame tube is the core part of reverse-flow combustors, which is typically fabricated from GH3536 superalloy, a solid solution nickel-based alloy with good mechanical properties, excellent oxidation and corrosion resistance [11, 12]. Under aero-engine service conditions, materials are routinely subjected to prolonged exposure to high pressure, elevated temperature, steep temperature gradient, and repeated start-stop and acceleration-deceleration cycles, which inevitably cause fatigue, creep and oxidation damage [13-15]. Corrosive gases (e.g., SO_2 , SO_3) present in combustion products can react with metal oxides or deposited salts on the flame tube's inner wall surface to form sulfate salts (e.g., Na_2SO_4 , CaSO_4), among which CaSO_4 may originate directly from impurity particles in the intake air. These sulfate salts act as corrosive media, leading to severe hot corrosion [16, 17]. In recent years, several studies have investigated the high-temperature properties of AM-fabricated GH3536 superalloy. For example, Ni et al. showed that compared to heat treatment (HT) samples, hot isostatic pressing-heat treatment (HIP-HT) samples exhibited significantly reduced porosity. At a strain amplitude of 0.20%, the HIP-HT samples had a markedly longer fatigue life [18]. Min et al. reported that L-PBF fabricated GH3536 superalloy exhibited superior oxidation resistance compared to the forged samples during a 500 h exposure at 950 °C. This enhancement was attributed to the cellular structure formed during L-PBF, which facilitated rapid Cr diffusion and promoted the early formation of a protective Cr_2O_3 layer [19]. Moreover, the corrosion resistance of L-PBF fabricated

GH3536 superalloy was significantly improved by solution and aging treatments, and was further enhanced with increasing aging temperature (750–950 °C) [20]. These preliminary investigations provide a basic understanding of L-PBF fabricated GH3536 superalloy at high temperature. However, systematic investigations into the failure of L-PBF fabricated GH3536 superalloy and its application in critical components—such as reverse-flow combustor flame tubes—under service conditions remain limited.

This paper reports the first study on the microstructural damage of an L-PBF fabricated GH3536 superalloy reverse-flow combustor flame tube tested under near-service conditions. The microstructures, crack propagation and corrosion behavior after the test are systematically investigated. The failure mechanisms of the flame tube under the combined effects of stress and corrosion are elucidated, providing important insights for the future optimization of flame tube design and maintenance.

2. Experimental

2.1. Additive manufacturing

GH3536 superalloy powders used for the L-PBF fabrication of the reverse-flow combustor flame tube had a particle size distribution of 15–53 μm , and the nominal composition of the alloy was (wt.%): C 0.10, Al 0.50, W 0.60, Mn 1.00, Si 1.00, Co 1.50, Mo 9.00, Fe 18.00, Cr 22.00, Ni balance. In the L-PBF process, the following printing parameters were used: laser power of 245 W, scan speed of 1198.6 mm/s, hatch spacing of 0.08 mm, and layer thickness of 0.04 mm. To mitigate typical L-PBF fabricated defects such as pores and microcracks, the as-printed flame tube underwent post-processing, including HIP (1170 °C, 140 MPa, 4 h), followed by HT (1177 °C, 90 min). The outer surface of the flame tube was subjected to coarse grinding, fine polishing and sandblasting to enhance surface quality. A structure diagram of the reverse-flow combustor and a top view of the flame tube's inner wall surface are shown in Fig. 1. Featuring a complex structure, the flame tube contains multiple types of holes, including primary holes, film cooling holes and dilution holes.

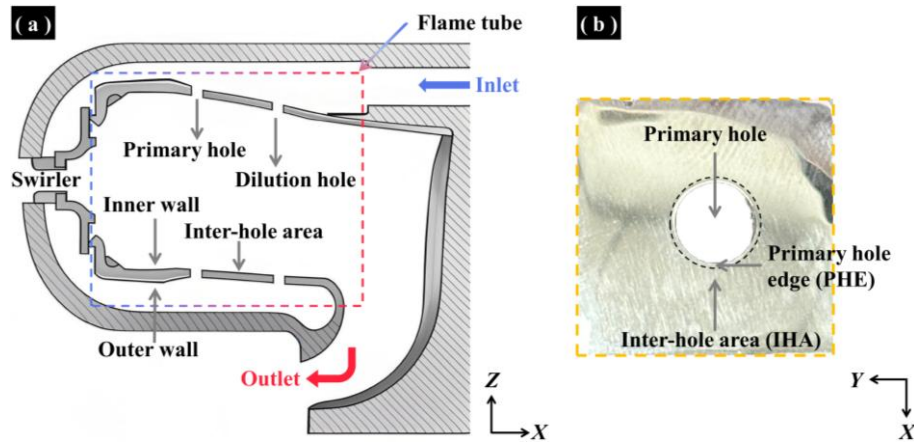


Fig. 1. Schematic diagram of the reverse-flow combustor and a top view of the flame tube inner wall surface.

2.2. Near-service combustion test

Subsequently, the near-service combustion test was conducted on the L-PBF fabricated flame tube to replicate the harsh operating conditions of actual combustors. The combustor inlet pressure and temperature were 1.4 MPa and 330 °C, respectively. During the subsequent peak-combustion stage, the system sustained a temperature rise exceeding 680 °C for 2.5 h, with the combustor outlet temperature exceeding 1010 °C. Heat-sensitive paint was applied to the flame tube's outer wall to calibrate the temperature distribution, indicating that the temperature of the outer wall in the main combustion zone reached approximately 900–1150 °C. Although the inner wall's temperature could not be measured directly, the heat transfer characteristics of the combustor suggested that its temperature was significantly higher.

After the test, some cracks were observed in different regions of the flame tube's inner wall surface, initiating from two adjacent regions: the primary hole edge and the inter-hole area. For further characterization, samples measuring 20.0 mm × 20.0 mm × 1.4 mm were taken from the failed flame tube's inner wall, using wire electrical discharge machining. For comparison, reference samples were fabricated under the same L-PBF and post-processing conditions but were not subjected to the near-service combustion test. As confirmed by SEM and XCT images (Fig. 2), the reference samples exhibited a dense, crack-free microstructure with negligible residual porosity.

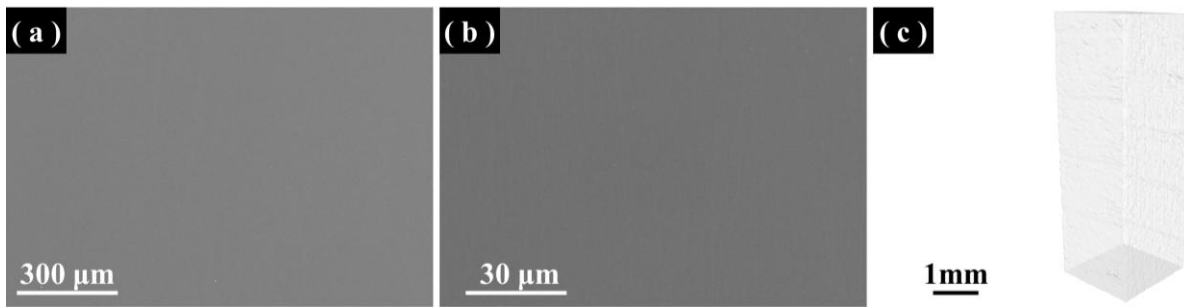


Fig. 2. SEM images and defect distribution of L-PBF fabricated GH3536 superalloy perpendicular to the building direction.

2.3. Microstructure characterizations

Samples for scanning electron microscopy (SEM) observation were prepared by mechanical grinding with SiC papers, followed by mechanical polishing and etching in a mixed solution (8 mL H₂O + 5 mL HNO₃ + 15 mL HCl) for 20 s. SEM, electron backscatter diffraction (EBSD) and energy-dispersive spectroscopy (EDS) characterizations were performed using an FEI Nova NanoSEM 450 SEM outfitted with EBSD and EDS detectors. X-ray computed tomography (XCT) of the reference samples (with dimensions of 2 mm × 2 mm × 10 mm and the longitudinal axis perpendicular to the build direction) was conducted using an XPloreVista 2000 system, with 1200 projections acquired at a tube voltage of 130 kV, an exposure time of 1 s and a voxel size of about 3.03 μm. Segmentation and analysis were performed using Avizo version 2020. X-ray diffraction (XRD, Panalytical X'Pert Pro) was employed to characterize the phase composition of corrosion products, utilizing a copper target ($\lambda = 1.5406 \text{ \AA}$) with a generator voltage of 40 kV and a tube current of 40 mA. The scanning range was $2\theta = 20^\circ\text{--}90^\circ$, and the scan-step size was 0.02° . X-ray photoelectron spectroscopy (XPS, Thermo ScientificTM NexsaTM) was used to analyze the elemental composition of the corrosive media. Transmission electron microscopy (TEM) specimens were prepared using both twin-jet electropolishing (TJE, TenuPol-5) and focused ion beam (FIB, Thermo Scientific Scios 2 DualBeam) systems. TEM characterization was performed using a spherical aberration-corrected FEI Titan G2 80-200 ChemiSTEM operating at 200 kV, equipped with a high-angle annular dark-field (HAADF) detector, super-X EDS and a bright-field (BF) detector.

3. Results

3.1. Cracking behavior on the inner wall surface

Since the primary holes experienced higher temperature exposure during the combustion test, microstructural damage mainly occurred in their vicinity. Cracks were visible to the naked eye

around the primary holes, and detailed characterizations further revealed cracks at various locations on the inner wall surface. Fig. 3(a) shows the overall morphology of a typical cracked region on the inner wall surface, which spanned more than 3000 μm in length. This cracked region comprises several cracks that initiated independently at the primary hole edge (PHE) and inter-hole area (IHA), propagating along the inner wall surface and tending to coalesce into a continuous crack. Fig. 3(b) shows a magnified view of the crack initiated from the primary hole edge in Fig. 3(a), where the primary crack propagated from the primary hole edge toward the inter-hole area, with several connected secondary cracks. Fig. 3(c–e) illustrates crack initiation in other regions. At the primary hole edge, multiple surface pits nucleated both at grain boundaries (GBs) and within grains, and most intragranular pits deepened and resulted in the initiation of transgranular cracks (Fig. 3(c) and (d)). In contrast, the cracks observed in the inter-hole area typically exhibited intergranular features, with a maximum crack width reaching up to 27 μm (Fig. 3(e)).

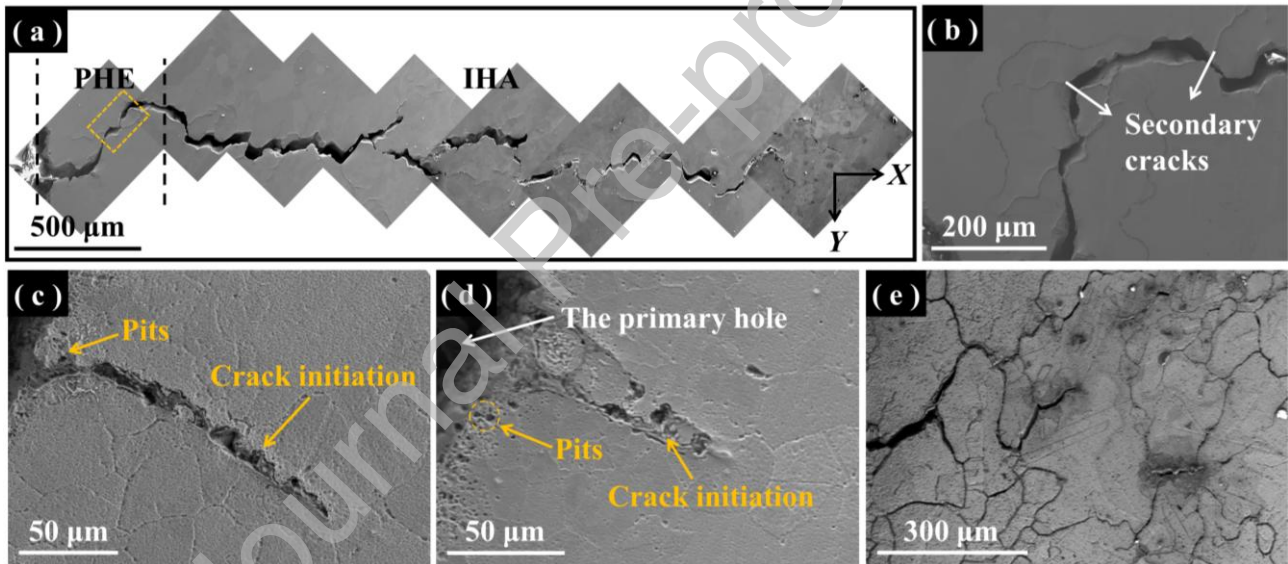


Fig. 3. SEM images of cracked regions on the inner wall surface of the flame tube: (a) a typical cracked region on the inner wall surface, consisting of several independently initiated cracks; (b) a magnified image of the area surrounded by the orange dotted box in Fig. 3(a); (c, d) transgranular crack initiation at the primary hole edge; (e) intergranular cracks in the inter-hole area.

To further investigate the crack behavior, EBSD analysis was conducted to elucidate the relationship between the cracking path and microstructural features. Fig. 4 presents the propagation path of the crack shown in Fig. 3(a). The inverse pole figure (IPF) map (Fig. 4(a)) and GB map (Fig. 4(b)) reveal that the main crack propagated predominantly along GBs, with partial transgranular features in some regions, indicating a GB-dominated fracture on the inner wall surface during the

test. In the IPF map, noticeable color gradients are identifiable within several grains, particularly on both sides of the crack. This indicates significant changes in intragranular orientation, implying that these regions underwent substructural evolution induced by cyclic loading during the fatigue process prior to cracking. Notably, some voids were observed at high-angle GBs (HAGBs), as indicated by the orange arrows in Fig. 4(b). According to previous studies, HAGBs act as preferential sites for void formation under high-temperature creep damage, and inclusions at HAGBs can further promote void nucleation [21]. Additionally, a high density of low-angle GBs (LAGBs) and significantly elevated kernel average misorientation (KAM) values were observed around the crack path (Fig. 4(b) and (c)). The LAGBs are sub-GBs formed by the accumulation and rearrangement of dislocations during fatigue-creep loading. The coexistence of dense LAGBs and high KAM values indicates localized accumulation of plastic strain and increased dislocation density on the inner wall surface during the test.

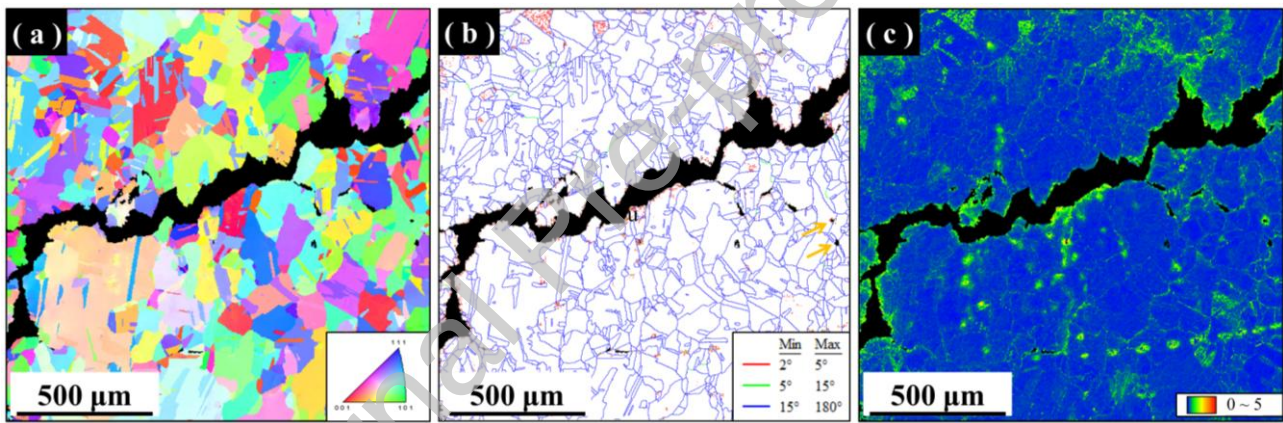


Fig. 4. EBSD analysis of the crack on the inner wall surface: (a) IPF map; (b) GB map; (c) KAM map.

TEM characterization was further carried out on the inner wall surface to investigate the local plastic deformation mechanisms and dislocation structures (Fig. 5). At the primary hole edge, dense dislocation networks and numerous slip bands were observed within the grains, accompanied by significant dislocation pile-ups and tangles (Fig. 5(a)). In the inter-hole area, dislocation loops and ordered dislocation arrangements were present within grains (Fig. 5(b)). Particularly, some junctions formed between twin boundaries (TBs), with a high density of dislocations accumulated at TBs (Fig. 5(c)). Theoretically, these TBs can act as barriers to dislocation motion, and the accumulation of dislocations at TBs under prolonged fatigue-creep loading may induce crack initiation.

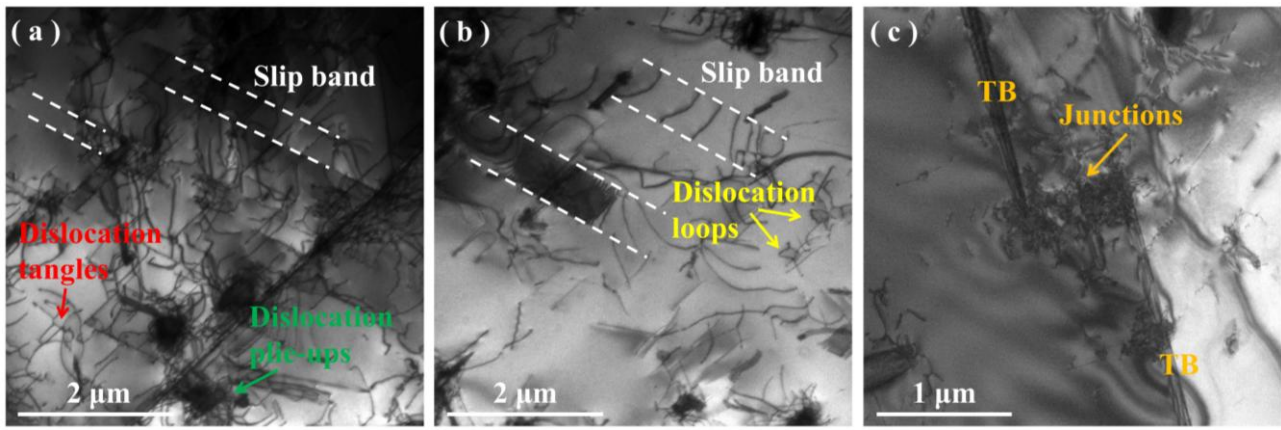


Fig. 5. TEM images of different regions on the inner wall surface: (a) primary hole edge ; (b) inter-hole area; (c) twin boundary junctions observed in the inter-hole area.

3.2. Corrosion behavior on the inner wall surface

In addition to the fatigue-creep damage, cracking on the inner wall surface was also strongly influenced by environmental degradation during the combustion test. Oxidation and hot corrosion can degrade the microstructural integrity and mechanical properties of the L-PBF fabricated GH3536 superalloy, thereby accelerating crack initiation and propagation.

XRD pattern of corrosion products on the inner wall surface is shown in Fig. 6. Results indicate that corrosion products primarily consist of Cr_2O_3 , Fe_2O_3 , NiCr_2O_4 , NiFe_2O_4 , Al_2O_3 and SiO_2 . In this pattern, no distinct diffraction peaks corresponding to NiO were detected, likely attributable to the further reaction of NiO with Fe_2O_3 and Cr_2O_3 to form more stable spinel phases (NiFe_2O_4 and NiCr_2O_4) [22].

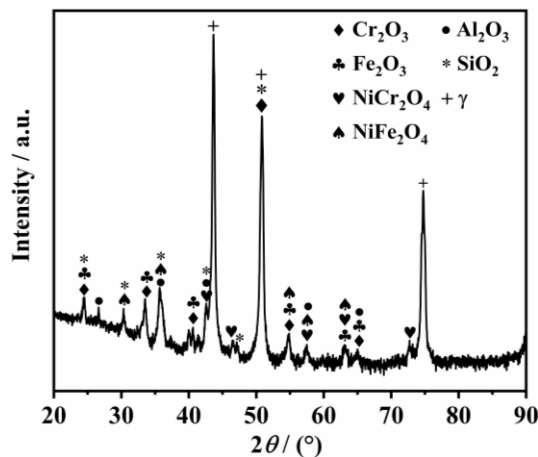


Fig. 6. XRD pattern of corrosion products on the inner wall surface of the flame tube.

Fig. 7 shows the cross-sectional morphology and corresponding elemental distributions of the

inner wall surface, revealing a highly heterogeneous corrosion layer. As shown in Fig. 7(a) and (c), the corrosion layer penetrated into the substrate, with a maximum thickness of approximately 20 μm observed in the region shown in Fig. 7(a). The nonuniform corrosion was likely caused by irregular gas flow and heterogeneous distribution of corrosive media. The corrosion layer displayed a porous and loose structure, characterized by internal voids. These voids were closely associated with the outward diffusion and depletion of metallic elements from the substrate and tended to grow as corrosion progressed. Combined EDS line scans and XRD analyses revealed that the outer region of the corrosion layer was primarily composed of Fe_2O_3 , while the inner region mainly consisted of Cr_2O_3 , with relatively low Ni content detected throughout the corrosion layer. Additionally, Al_2O_3 and SiO_2 were identified above the Cr_2O_3 region (Fig. 7(a) and (b)). As shown in Fig. 7(d) and (h), S from the corrosive media penetrated through the porous Fe_2O_3 layer, leading to the formation of Cr-rich sulfide.

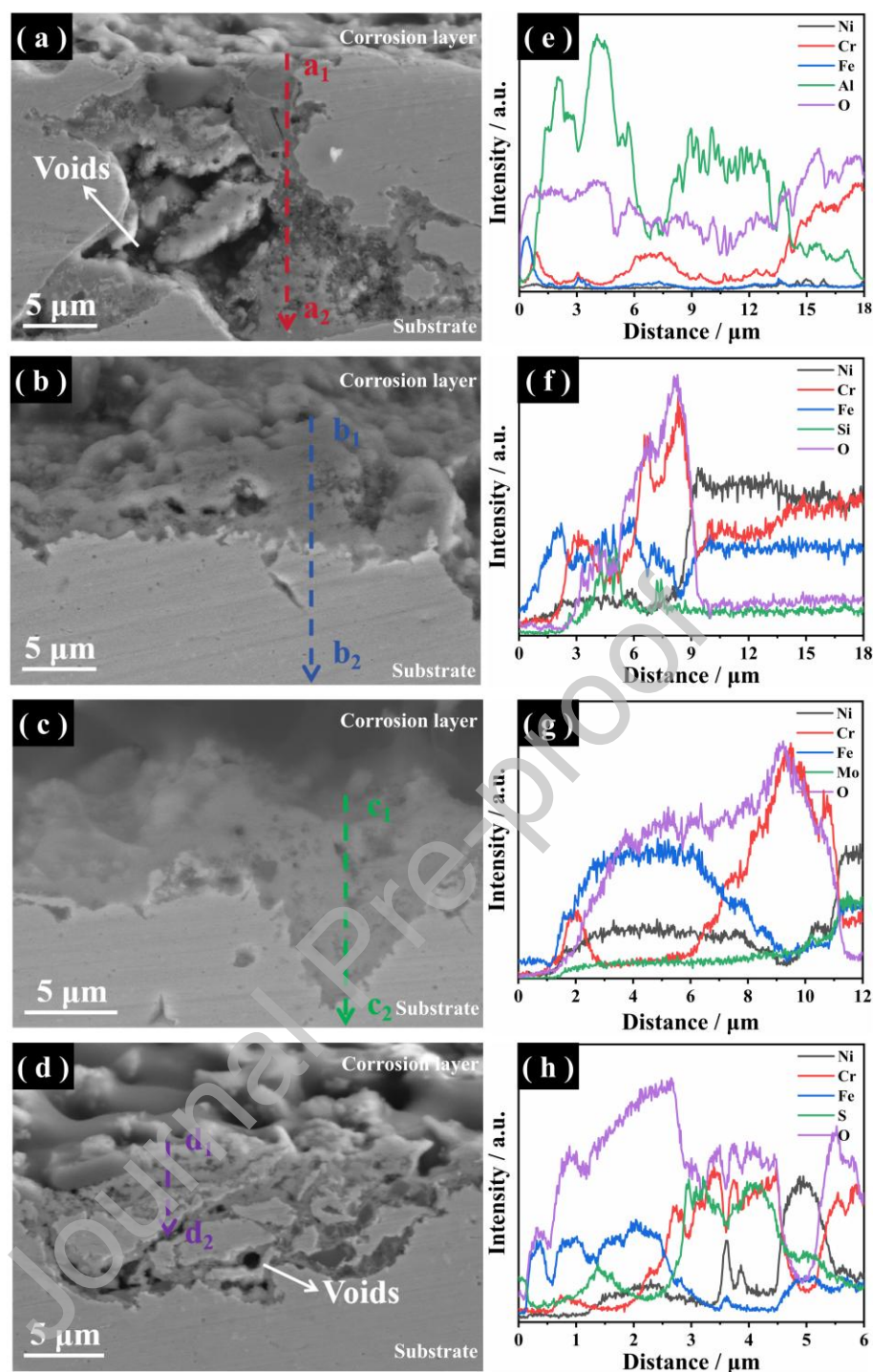


Fig. 7. Cross-sectional microstructure of the inner wall surface: (a–d) SEM images of the corrosion layer; EDS analysis of the (e) a_1a_2 line, (f) b_1b_2 line, (g) c_1c_2 line, (h) d_1d_2 line. The line scan direction in (e–h) is from the corrosion layer toward the substrate.

To confirm the presence of sulfate species contributing to hot corrosion, XPS analysis was performed on the corrosion layer scraped from the inner wall surface, with the fitted spectra shown in Fig. 8. The Fe 2p spectrum (Fig. 8(a)) can be deconvoluted into two components at 709.78 and 711.23 eV, corresponding to Fe^{2+} and Fe^{3+} species, respectively. The Cr 2p spectrum (Fig. 8(b))

exhibits peaks at 570.33, 575.88 and 585.73 eV, characteristic of Cr_2O_3 and NiCr_2O_4 . Consistently, the O 1s spectrum (Fig. 8(c)) shows a peak at 529.58 eV, characteristic of lattice oxygen in metal oxides (M–O) such as Fe_2O_3 (530.2 eV) and Cr_2O_3 (529.9 eV). The S 2p spectrum (Fig. 8(d)) displays two peaks at 163.58 and 168.98 eV, attributable to S^{2-} in sulfides and S^{6+} in sulfate species, respectively; whereas the Ca 2p and Na 1s spectra (Fig. 8(e) and (f)) exhibited binding energies consistent with Ca^{2+} and Na^+ in sulfate species [23]. These results strongly indicate the presence of Na_2SO_4 and CaSO_4 on the inner wall surface.

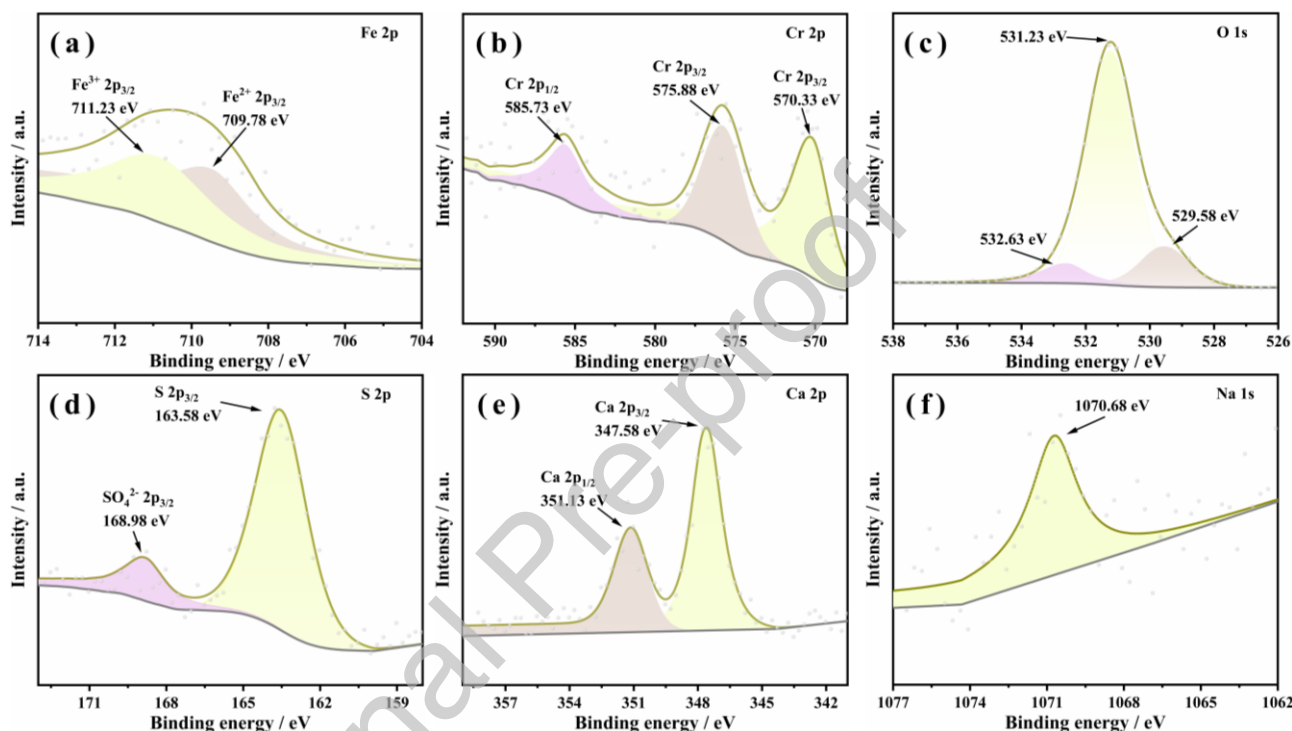


Fig. 8. XPS fitting results of the corrosion layer scraped from the inner wall surface: (a) Fe 2p; (b) Cr 2p; (c) O 1s; (d) S 2p; (e) Ca 2p; (f) Na 1s.

Numerous pits appeared on the inner wall surface after the combustion test (Fig. 3(c) and (d)). However, SEM and XCT analyses of the reference samples confirmed a dense, pit-free microstructure, indicating that these pits were formed during the test rather than being L-PBF fabricated defects. Fig. 9(a–c) shows the cross-sectional morphology of pits on the inner wall surface and the corresponding EDS analyses. The pits were open to the inner wall surface, extended into the substrate, and were surrounded by a distinct corrosion layer. At Point 1 in Fig. 9(a), trace amounts of Ca and S were detected in the pit, and an EDS line scan across the pit revealed pronounced S and Ca peaks (Fig. 9(b)), indicating the enrichment of CaSO_4 . At Point 2 in Fig. 9(c), the mass fractions of Ca, Na and S in the corrosion layer were 5.05%, 1.52% and 0.74%,

respectively. Combined with XPS results, these observations support the inference that Na_2SO_4 and CaSO_4 were involved in hot corrosion and contributed to the formation of corrosion pits. By removing the corrosion layer from the inner wall surface, the walls of both main and secondary cracks were found to be covered with oxides, primarily composed of Cr_2O_3 and minor Al_2O_3 , as shown in Fig. 9(d). At the primary hole edge, some corrosion pits induced by hot corrosion were distributed nonuniformly along the main crack (Fig. 9(e)). Furthermore, in the inter-hole area, larger and denser corrosion pits were observed along both sides of the GBs, with partial GB cracking (Fig. 9(f) and (g)). This is likely associated with the formation of GB corrosion products.

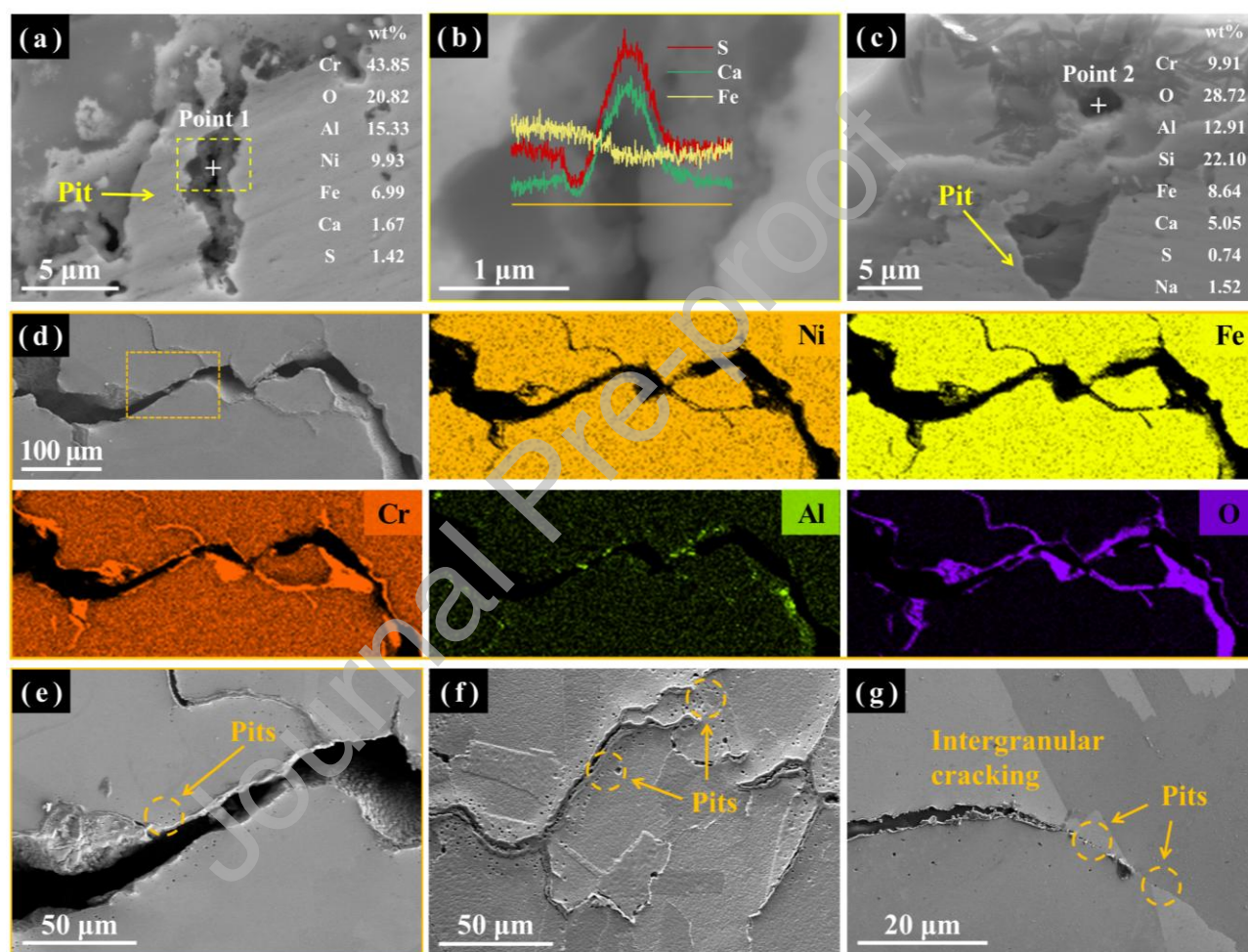


Fig. 9. Microstructure of corrosion pits and cracks on the flame tube inner wall surface: (a) cross-sectional morphology of a pit and EDS point analysis at Point 1; (b) enlarged view of the yellow dashed box in Fig. 9(a) with EDS line scan; (c) cross-sectional morphology of another pit and EDS point analysis at Point 2; (d) SEM image and corresponding EDS mapping of cracks at the primary hole edge; (e) enlarged view of the main crack in Fig. 9(d); (f, g) SEM images of partially cracked grain boundaries in the inter-hole area.

As shown in Fig. 10(a), significant internal oxidation occurred beneath the corrosion layer in the inter-hole area. Strip-like Cr_2O_3 precipitates were continuously observed along GBs, with widths of 0.18–0.55 μm , and corrosion pits preferentially formed in the vicinity of these oxidized GBs. In contrast, the reference sample exhibited intact GB structures without any noticeable oxygen-rich precipitates, and metallic elements such as Ni, Fe, Cr and Mo were uniformly distributed (Fig. 10(b)). This difference indicates that fatigue-creep and high-temperature exposure during the combustion test can promote the intergranular oxidation and significantly reduce the GB strength, consistent with the in-situ TEM study by Zhou et al., where thermal-stress exposure can accelerate the GB oxidation and thereby GB cracking [24].

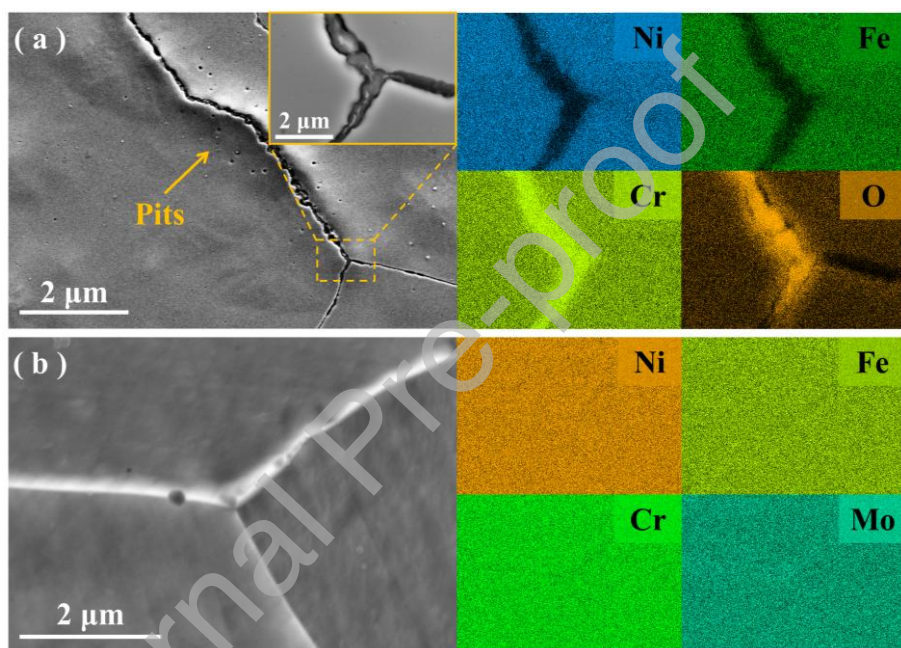


Fig. 10. Comparison of grain boundary microstructures between the inter-hole area and reference sample: SEM images and corresponding EDS mapping images of (a) the inter-hole area and (b) the reference sample.

Further TEM analysis also revealed the intergranular sulfidation in the specimens extracted from GBs adjacent to the crack path in the inter-hole area. As shown in Fig. 11(a), a Cr-rich sulfide precipitate was observed along GBs, exhibiting a band-like morphology with a width of approximately 75 nm. High-resolution HAADF-STEM images (Fig. 11(b) and (c)) and fast fourier transform (FFT, Fig. 11(d) and (e)) patterns further identified the precipitate as the cubic $\text{Cr}_4\text{FeNiS}_8$ phase (space group $\text{F}\bar{4}3\text{m}$, $a = b = c = 9.80$, $\alpha = \beta = \gamma = 90^\circ$). This phase preferentially precipitated along GBs and exhibited a well-defined orientation relationship with the γ matrix: $[011]_\gamma//$

$[110]_{\text{Cr}_4\text{FeNiS}_8}$ and $[001]_\gamma // [010]_{\text{Cr}_4\text{FeNiS}_8}$. Fig. 11(f) and (g) display the atomic structure of the $\text{Cr}_4\text{FeNiS}_8$ phase. The presence of brittle sulfides along GBs can weaken the GB cohesion and facilitate the microcrack initiation.

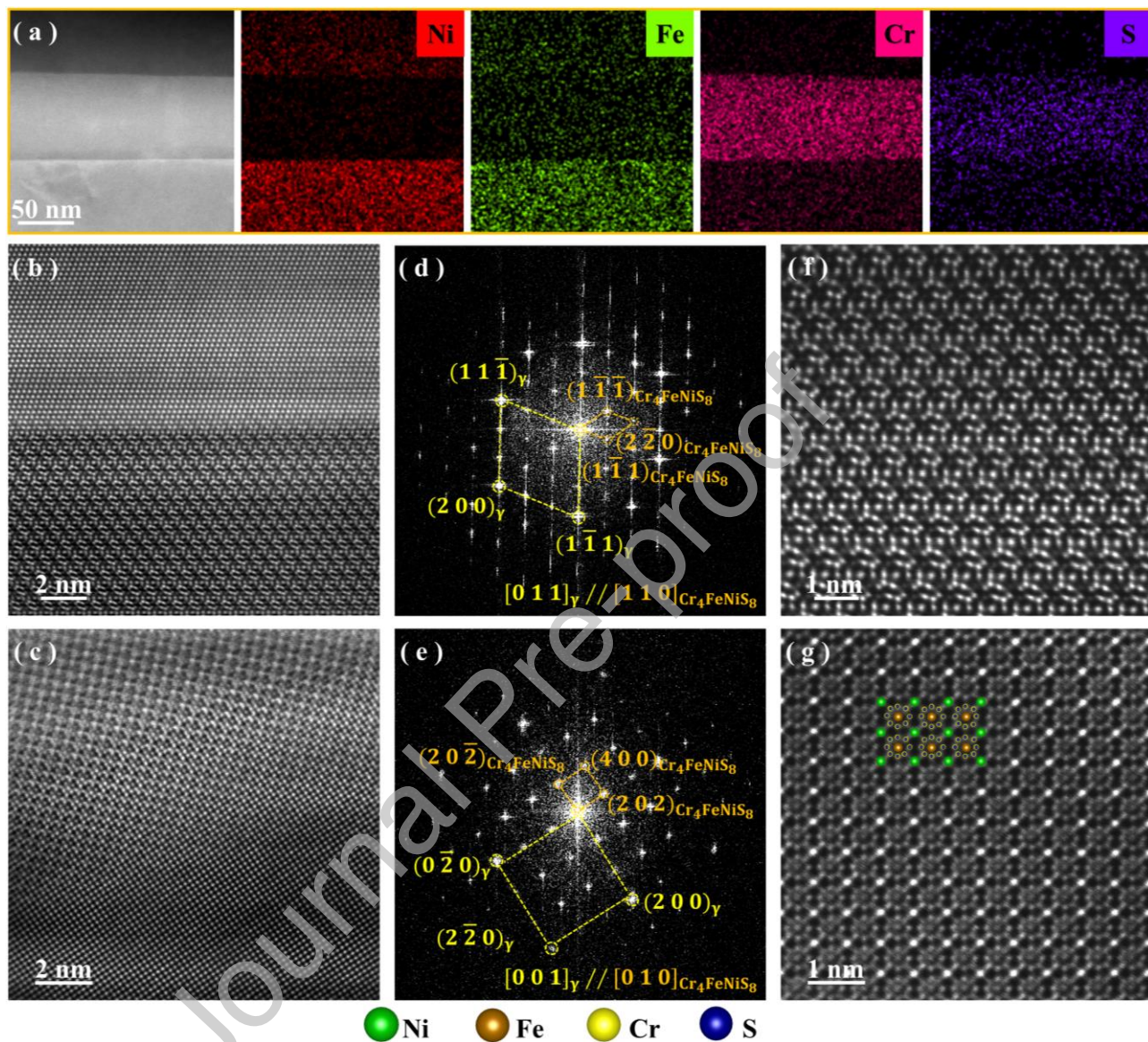


Fig. 11. Microstructural and atomic-scale characterization of the Cr-rich sulfide at the grain boundary: (a) HAADF-STEM image showing the morphology of the Cr-rich sulfide at the GB and the corresponding EDS mapping images; (b, c) high-resolution HAADF-STEM images of the $\text{Cr}_4\text{FeNiS}_8/\gamma$ interface along the $[011]_\gamma$ and $[001]_\gamma$ zone axes; (d, e) FFT patterns corresponding to Fig. 11(b) and (c); (f, g) atomic structure of $\text{Cr}_4\text{FeNiS}_8$ along the $[110]_{\text{Cr}_4\text{FeNiS}_8}$ and $[010]_{\text{Cr}_4\text{FeNiS}_8}$ zone axes.

To verify whether impurity elements such as O and S had segregated at GBs before the

combustion test, a representative general GB region (i.e., randomly selected, high angle, asymmetric) in the reference sample was characterized using HAADF-STEM and EDS, as shown in Fig. 12. Fig. 12(a) and (b) shows the microstructure of the representative general GB and its magnified view, respectively; Fig. 12(b) clearly resolves the atomic arrangement on one side of the GB. Fig. 12(c) and (d) presents the FFT patterns of the two adjacent grains, further confirming the difference in crystallographic orientation. Fig. 12(e) shows the EDS mapping of the region presented in Fig. 12(b), where no segregation of O, S or other impurity elements, nor any noticeable enrichment of substrate elements at the GB was observed. Based on Figs. 10 and 12, it can be concluded that the GB regions prior to the combustion test exhibited a uniform microstructure and chemical distribution, without apparent impurity segregation.

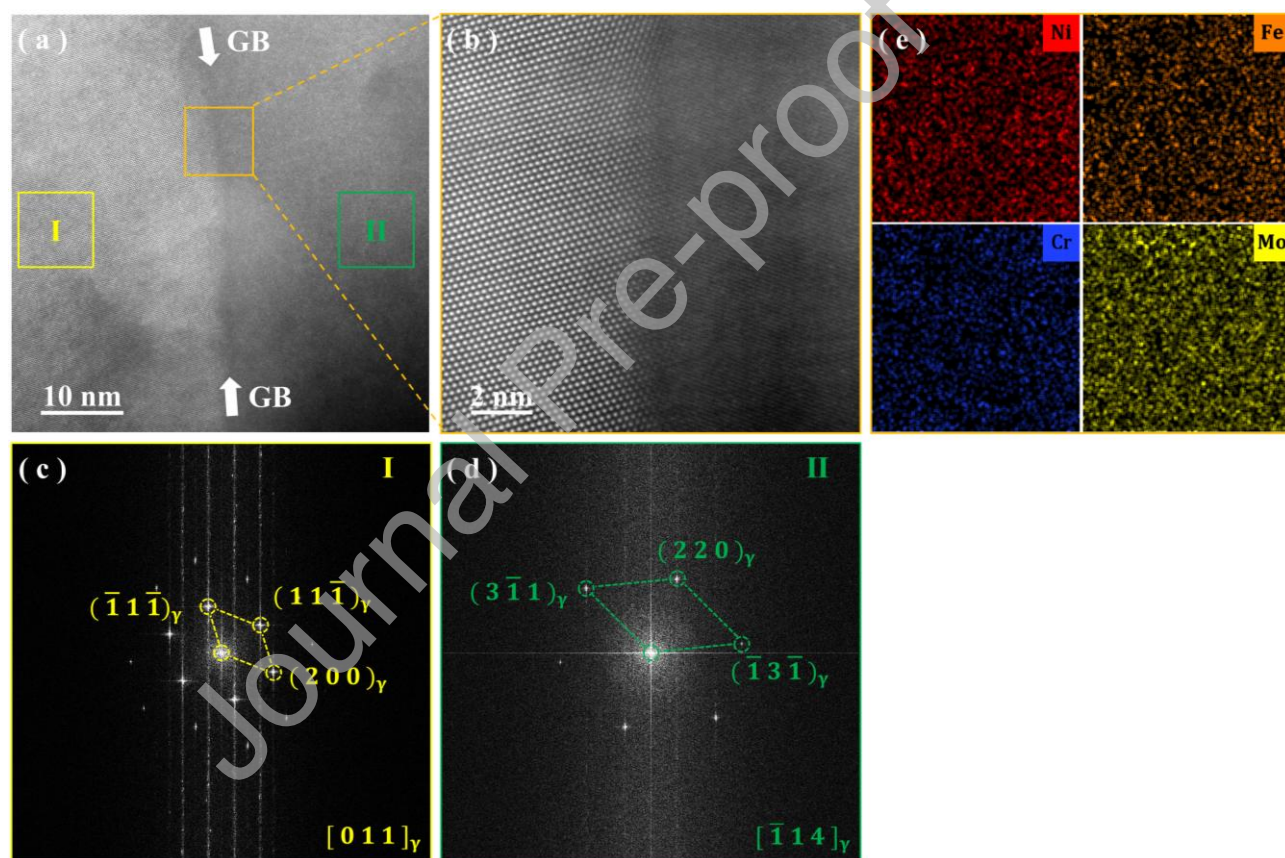


Fig. 12. A representative general GB region (i.e., randomly selected, high-angle, asymmetric) in the reference sample: (a) HAADF-STEM image of the representative general GB; (b) high-resolution HAADF-STEM image of the area surrounded by the orange dotted box in Fig. 12(a); (c, d) FFT patterns corresponding to the yellow and green dotted box areas in Fig. 12(a); (e) EDS mapping images of the region shown in Fig. 12(b).

4. Discussion

To better elucidate the failure mechanisms of the L-PBF fabricated GH3536 flame tube during the combustion test, Fig. 13 presents a schematic overview of the failure processes. Fig. 13(a) and (b) illustrates the cross-sectional corrosion behavior on the inner wall surface, while Fig. 13(c) and (d) depicts crack initiation and propagation along the inner wall surface.

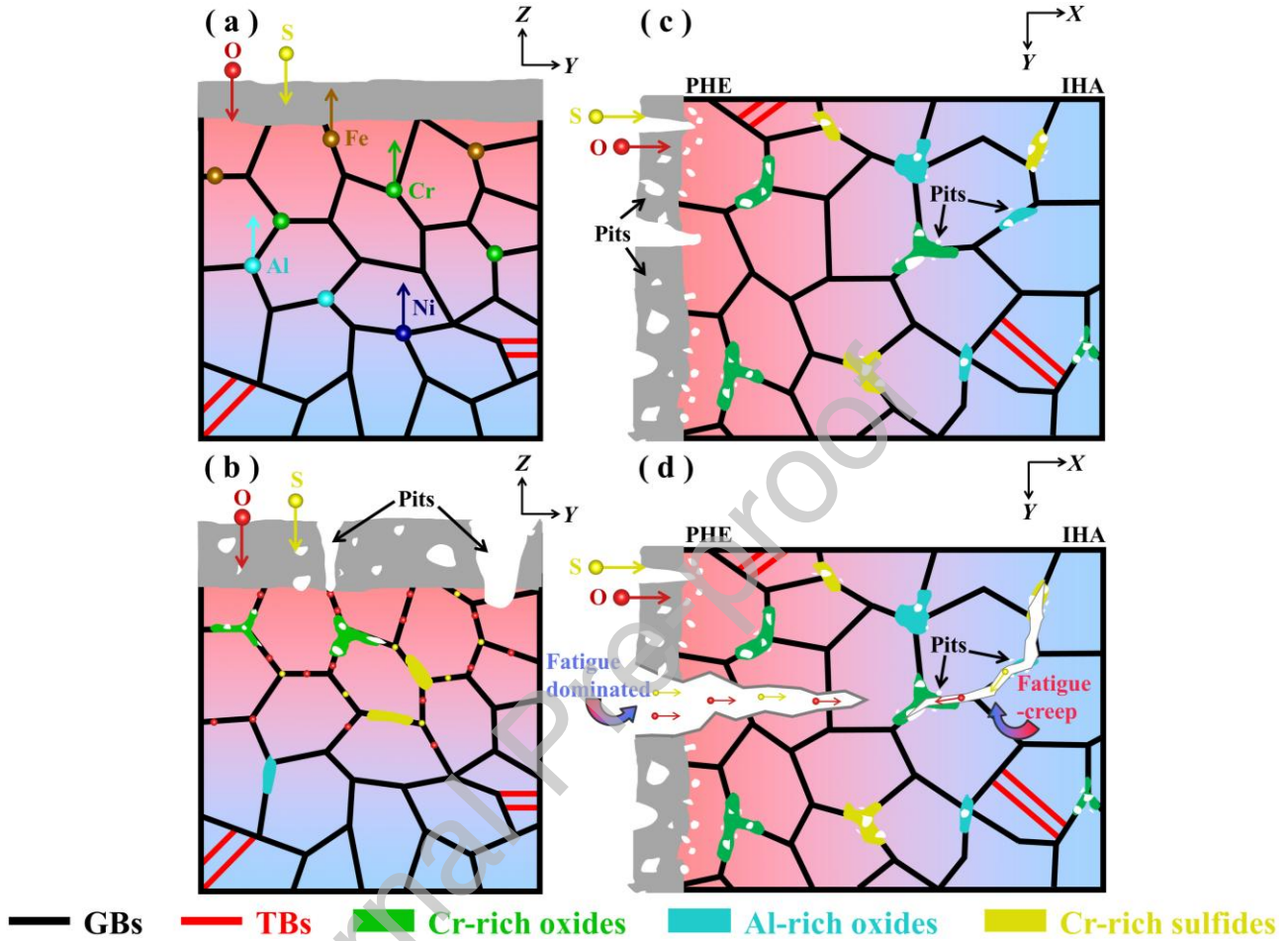


Fig. 13. Schematic of the failure processes of the reverse-flow combustor flame tube.

4.1. Failure of the protective oxide layer

During the initial stage of the combustion test, oxidation preferentially occurred on the inner wall surface, where Ni, Fe, Cr, and Al diffused outward at different rates to form a protective oxide layer (Fig. 13(a)). However, as shown in Fig. 7, the integrity and protective function of the oxide layer were degraded significantly due to two concurrent processes:

1. Stress-induced damage: the oxide layer consists of several stratified phases with different thermal expansion coefficients, both relative to each other and to the substrate. Under fatigue-creep loading, these mismatches induce stress concentrations within the oxide layer and at the oxide-substrate interface [25, 26]. Additionally, the brittleness of the oxide layer, coupled with intrinsic microvoids and microcracks (Fig. 7(a) and (d)), further promotes fracture and

spallation of the oxide layer, creating diffusion paths for O and S [25, 27, 28].

2. Corrosion-induced damage: Corrosive media and corrosive gases can locally dissolve the oxide layer, allowing S to diffuse through the compromised oxide layer and react with the substrate to form sulfides (Fig. 7(h)). Under high oxygen partial pressure, some sulfides may undergo reoxidation and release S. The released S can further diffuse inward, establishing a sulfidation-oxidation-resulfidation cycle that disrupts the continuity of the oxide layer and accelerates substrate corrosion [29, 30].

4.2. GBs served as preferential paths for oxidation and sulfidation

As shown in Figs. 10 and 11, Cr_2O_3 and $\text{Cr}_4\text{FeNiS}_8$ precipitated along GBs beneath the corrosion layer. Similar to the protective oxide layer on the inner wall surface, Cr_2O_3 along GBs exhibited partial spallation due to the combined effects of stress and corrosion (Fig. 10(a)), which compromised its integrity and facilitated the further diffusion of O and S into the substrate. Following the observation of Cr_2O_3 and $\text{Cr}_4\text{FeNiS}_8$, the inter-hole area was ground with SiC paper to expose deeper regions, from which TEM specimens were prepared. As shown in Fig. 14, lamellar Al_2O_3 formed preferentially along deeper GBs, while Cr was only locally enriched around Al_2O_3 without extensive formation of Cr_2O_3 . This can be attributed to the decrease in oxygen partial pressure as O continues to diffuse into the substrate. Under such conditions, Al exhibits a stronger oxygen affinity than Cr and thus facilitates the formation of Al_2O_3 [31].

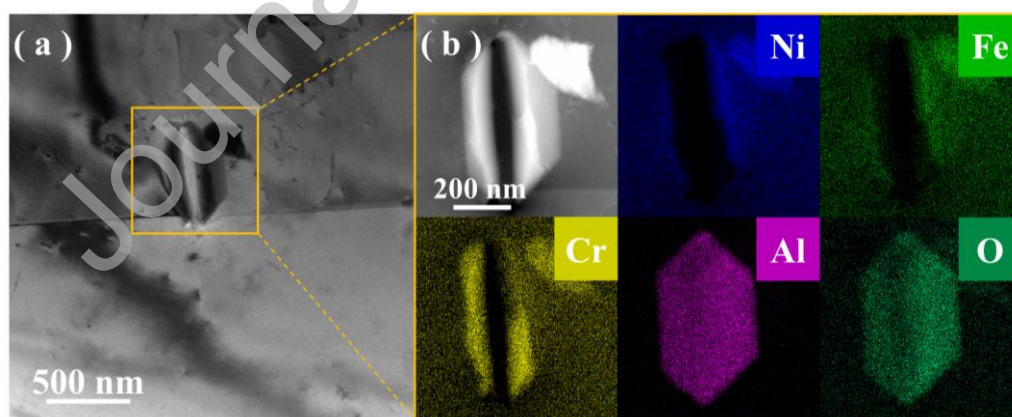


Fig. 14. (a) TEM image of Al_2O_3 at deeper grain boundary regions within the substrate; (b) corresponding EDS mapping images.

Theoretically, the loose-packed atoms and high excess volume of GBs naturally endow fast mass transport along the GBs. Under fatigue-creep loading, plastic deformation further increases defect density near GBs, thereby enhancing diffusion along GBs, and the GB diffusion coefficient is

much higher than that within the grain interior [32, 33]. The precipitation of Cr_2O_3 , Al_2O_3 , and $\text{Cr}_4\text{FeNiS}_8$ along GBs beneath the corrosion layer indicates that GBs are favorable nucleation sites for corrosion products. During the combustion test, GBs not only accelerated the outward diffusion of substrate elements but also provided effective paths for the inward diffusion of O and S, resulting in preferential oxidation and sulfidation along GBs, as illustrated in Fig. 13(b).

4.3. Initiation and propagation of corrosion pits and cracks

Notably, although both the primary hole edge and the inter-hole area lay within the highest-temperature region, they exhibited distinct crack morphologies due to differences in the local stress states and corrosion behavior. The primary hole edge experienced a higher temperature gradient and more severe fatigue damage than the inter-hole area, as evidenced by its higher dislocation density (Fig. 5). Additionally, severe turbulence generated by the intersection of high-temperature gas with cooling air at the primary hole edge stripped the oxide layer, while fatigue further aggravated this spallation, enabling O_2 and corrosive media to attack the exposed substrate and resulting in the formation of corrosion pits (Figs. 3(c), 3(d) and 9(e)).

Regarding the formation of corrosion pits, previous studies have mainly focused on hot corrosion dominated by Na_2SO_4 . In this study, CaSO_4 was also detected in the corrosion layer, with a higher content than Na_2SO_4 (Fig. 9(a) and (c)). This aligns with the composition of corrosive deposits found on post-service combustors, where CaSO_4 can largely replace Na_2SO_4 in driving hot corrosion [16, 17, 34]. Therefore, a hot corrosion mechanism dominated by CaSO_4 is briefly discussed here. Turbulence and local flow field variations resulted in nonuniform deposition of corrosive media on the inner wall surface, creating favorable sites for nonuniform corrosion and corrosion pit formation. Although the inner wall's service temperature was higher than that of the outer wall (900–1150 °C), it is uncertain whether it exceeded the melting point of pure CaSO_4 (1507 °C) [35]. Nevertheless, previous studies have shown that even in the unmelted state, CaSO_4 itself can damage the Cr_2O_3 layer and release SO_3 at about 750 °C [17]. When mixed with Al_2O_3 and SiO_2 , CaSO_4 undergoes accelerated decomposition and promotes oxidation and sulfidation by forming low-melting-point salts [35, 36]. The enrichment of Al_2O_3 and SiO_2 within and around the corrosion pits suggests that CaSO_4 , together with silicate and oxide deposits, plays an important role in oxide degradation and corrosion pits formation (Fig. 9(a) and (c)).

As corrosion pits deepened, stress concentrations intensified at their bottoms due to geometric effects and local hardness reduction, so that under fatigue-dominated loading, these intragranular

pits acted as preferential crack initiation sites and eventually evolved into transgranular cracks [37-40]. Therefore, at the primary hole edge, crack initiation was mainly governed by fatigue damage concentrated at intragranular pits, ultimately resulting in transgranular cracks (Fig. 13(c) and (d)).

In contrast, the inter-hole area experienced weaker turbulence and lower fatigue damage, making the corrosive media and oxide layer on the surface less prone to removal and spallation. As discussed in Section 4.2, GBs in the inter-hole area acted as preferential nucleation sites for corrosion products. These corrosion products along GBs can not only cause local Cr depletion and oxide layer weakening but also induce microcracks and voids along GBs under fatigue-creep loading [41]. Consequently, the oxide layer and underlying substrate lost effective support and became more susceptible to local collapse and spallation under combined stress and corrosion damage, resulting in the formation of corrosion pits along GBs (Fig. 10(a)). Moreover, multiple corrosion pits were distributed ahead of the tip of the intergranular crack, indicating that corrosion pits facilitated the propagation of intergranular cracks (Fig. 9(g)). Thus, in the inter-hole area, cracks tended to nucleate and propagate preferentially along these weakened GBs under fatigue-creep loading, giving rise to severe intergranular cracking (Fig. 13(c) and (d)).

Once transgranular cracks initiated at the primary hole edge (Fig. 3(c) and (d)), the area of the substrate exposed to oxygen and corrosive media increased, thereby further accelerating substrate degradation. Under fatigue-dominated loading, the crack walls collapsed, and the crack tips were driven to propagate longitudinally, assisted by stress-assisted GB oxidation (SAGBO) or dynamic embrittlement (DE) [27, 42-44]. As these cracks propagated into the inter-hole area, they tended to deflect onto weakened GBs and coalesce with intergranular cracks nucleated in the inter-hole area, forming large cracks with mixed fracture morphologies (Fig. 3(a)).

4.4. Corrosion susceptibility and mitigation strategies for AM-fabricated alloys

Compared with conventionally manufactured alloys, AM-fabricated alloys are more susceptible to pitting corrosion, intergranular oxidation and sulfidation, owing to inherent printing defects such as porosity, surface roughness and a reduced fraction of special GBs [45-47]. Porosity provides preferential sites for oxide layer rupture and corrosion pit initiation, while surface roughness promotes the accumulation of corrosive media in surface depressions, thereby accelerating hot corrosion [47]. In addition, the reduced fraction of special GBs weakens intergranular corrosion resistance, while residual stresses and dislocation pile-ups generated during

the AM process often concentrate at GBs, rendering them more active diffusion paths [46]. Although HIP and sandblasting can mitigate some of these issues, corrosion pits, intergranular oxidation and sulfidation were still observed in the L-PBF fabricated GH3536 flame tube in this study, highlighting the intrinsic corrosion susceptibility of AM-fabricated alloys. Therefore, further strategies such as GB engineering and microalloying are essential to enhance their corrosion resistance and service reliability. For example, high-temperature annealing of L-PBF fabricated Inconel 718 can activate residual stress-driven recrystallization, significantly increasing the fraction of $\Sigma 3$ TBs [48], and adding only 0.01% Y to Inconel 825 can markedly improve the resistance to pitting and intergranular corrosion [49]. It is also noteworthy that our flame tube was tested in a bare substrate condition without any protective coating, unlike combustors in actual service. The application of a protective coating may help to enhance the damage resistance. Overall, the results of this study provide important insights into the degradation and failure of the GH3536 flame tube under service conditions, which will benefit the optimization of other AM-fabricated superalloy parts for harsh applications.

5. Conclusions

This paper systematically investigated the failure mechanisms of a GH3536 superalloy reverse-flow combustor flame tube fabricated by L-PBF, under near-service combustion conditions. The main conclusions are as follows:

(1) Cracking of the inner wall surface occurs mainly in an intergranular mode with partial transgranular features. EBSD and TEM analyses revealed localized plastic strain accumulation along crack paths, characterized by sub-GBs, high KAM values, dense dislocation networks and slip bands.

(2) The corrosion layer on the inner wall surface exhibits a porous and heterogeneous structure, containing Fe_2O_3 , Cr_2O_3 , Al_2O_3 , SiO_2 and Cr-rich sulfides. XPS and EDS analyses confirmed the presence of CaSO_4 and Na_2SO_4 , which act as corrosive media contributing to hot corrosion.

(3) Before the combustion test, no impurity segregation was observed at the GBs of L-PBF fabricated GH3536 superalloy. However, after exposure, the protective oxide layer on the inner wall surface degrades under the combined effects of stress and corrosion, allowing inward penetration of O and S. GBs serve as preferential diffusion paths, leading to intergranular oxidation and sulfidation.

(4) At the primary hole edge, fatigue causes intragranular pits to initiate transgranular cracks,

which propagate intergranularly toward the inter-hole area. In the inter-hole area, corrosion products and pits along GBs facilitate the initiation and growth of intergranular cracks. Overall, the failure of the flame tube is dominated by the synergistic effects of fatigue-creep, oxidation and hot corrosion.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yuxuan Duan: Data curation, Formal analysis, Investigation, Writing original - draft. **Haohan Ni:** Formal analysis, Investigation, Methodology. **Lei Qian:** Formal analysis, Methodology. **Xiaohong Shao:** Writing - review & editing, Methodology. **Qi Zeng:** Supervision, Funding acquisition, Writing - review & editing. **Huajie Yang:** Supervision, Funding acquisition, Writing - review & editing. **Jiangwei Wang:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing - review & editing

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Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in

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